LEARNER GUIDE

Function notation, domain and range

Algebraic Expressions and Identities · Lesson 8. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **function** is a rule that turns each input into exactly one output, like a machine. We write f(x) = 2x – 1, and f(3) means the output when the input is 3: put 3 where x was, so f(3) = 5. The inputs a function can take make its **domain**. Some rules cannot take certain inputs: 1/(x – 2) cannot take 2, because you cannot divide by zero, so 2 is **excluded**. The outputs that actually come out make its **range**.

What you are learning to notice

- Whatever is in the brackets replaces x: f(a) = 2a – 1.
- Solving f(x) = 9 means finding the input that gives the output 9.
- Two inputs may share an output (g(4) = g(–4) = 19), but one input never has two outputs.
- Exclude any input that would divide by zero or take the square root of a negative number.
- Some outputs never appear: x^2 + 3 is never less than 3, because $x^2 \geq 0$.

Moving through it, one step at a time

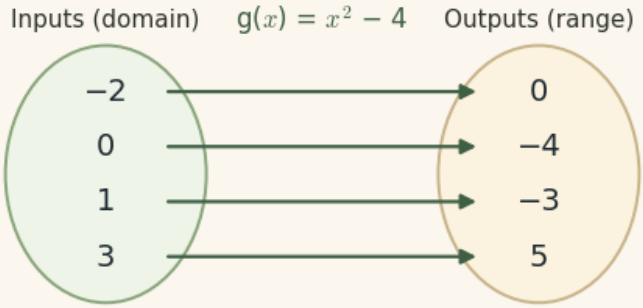
To evaluate, put the input where x was. To solve f(x) = k, write the equation and solve it. For the domain, ask: which inputs would jam the machine? For the range, draw a mapping diagram (or a graph) and ask: which outputs come out, and which can never appear?

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A mapping diagram and its range

g(x) = x^2 – 4 with domain {–2, 0, 1, 3}.

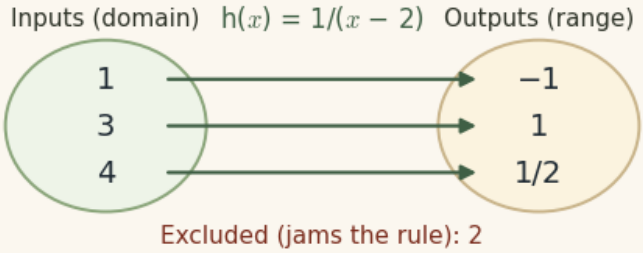


1. Feed in –2.	g(–2) = (–2)^2 – 4 = 0
2. Feed in 0 and 1.	g(0) = –4, g(1) = –3
3. Feed in 3.	g(3) = 9 – 4 = 5
4. Collect the outputs: the range.	{–4, –3, 0, 5}

Range: {–4, –3, 0, 5}. Each input has exactly one arrow, so g is a function.

Example 2 — An input that jams the machine

h(x) = 1/(x – 2). Which input must be excluded? Can h(x) equal 0?



1. Feed in 1, 3 and 4.	h(1) = –1, h(3) = 1, h(4) = 1/2
2. Feed in 2: you cannot divide by zero.	1/(2 – 2) = 1/0
3. So exclude 2 from the domain.	domain: all numbers except 2
4. If 1/(x – 2) = 0, multiply by (x – 2).	1 = 0, which is false
5. So 0 never comes out.	0 is not in the range

Domain: all numbers except 2. The output 0 can never appear.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (substitution, solving a linear equation, and knowing that you cannot divide by zero or take the square root of a negative number), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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